

DeployTree – Build & Compilation Management Dashboard Project Report

B.Tech CSE Semester Project

Student Details

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Course: B.Tech CSE

Project: DeployTree – Build & Compilation Management Dashboard

Problem Statement

Develop a frontend-only dashboard that simulates software project management, build execution, compilation monitoring, testing workflows, and deployment operations using Data Structures and Algorithms.

Tech Stack

React.js, JavaScript, CSS, React Hooks, Vite.

Component Architecture

App.jsx (Root) → Sidebar, Header, DashboardHome, CodeStructureViewer, BuildErrorHistory, BuildTaskQueue, FileLinkChecker, FileSizeSorter, CodeLinkHub, QuickestBuildTimeFinder, MockCPUWorkloadBalancer.

Features

1. Code Structure Viewer (Tree/DFS)
2. Build Error History (Stack)
3. Build Task Queue (Priority Queue)
4. File Link Checker (Hash Table)
5. File Size Sorter (Merge Sort)
6. Code Link Hub (Graph)
7. Quickest Build Finder (Dijkstra)
8. CPU Workload Balancer (Min Heap)

Additional: Dark Mode, Search, Toasts, Responsive UI, Charts.

Screenshots

Insert screenshots here:

- Dashboard Home
- Build Error History
- Build Task Queue
- File Size Sorter
- Graph Visualization
- CPU Workload Balancer

GitHub & Live Link

GitHub Repository: _____

Live Deployment (Vercel): _____

Conclusion

DeployTree demonstrates practical implementation of multiple Data Structures and Algorithms inside a modern React application. The project provides an interactive environment for understanding software build systems while improving frontend development and problem-solving

skills.